

Instructions

This mock mirrors the Infosys SP/DSE online assessment structure. Attempt it in one timed sitting; review using the highlighted answers.

Section 1 - Logical Reasoning | Section 2 - Quantitative Ability | Section 3 - Verbal Ability | Section 4 - Pseudocode/Technical | Section 5 - Coding (2 problems).

This is Mock Test 2 of 3; each mock uses a different question set.

Section 1 - Logical Reasoning

Q1. Some pens are books. All books are papers. Conclusion: Some pens are papers.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) None

Answer: (A) Follows

Authenticity: Frequently Repeated

Q2. Some cars are buses. Conclusion: Some buses are cars.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) None

Answer: (A) Follows

Authenticity: Frequently Repeated

Q3. Next term: 5, 10, 17, 26, ?

(A) 35

(B) 37 ✓

(C) 36

(D) 40

Answer: (B) 37

Authenticity: Frequently Repeated

Q4. Letter series: B, D, G, K, ?

(A) N

(B) O

(C) P ✓

(D) M

Answer: (C) P

Authenticity: Community Reported

Q5. If FRIEND is HTKGPF (each +2), then CANDLE is:

(A) ECPFNG ✓

(B) ECQFNG

(C) DBOEMF

(D) ECPGNF

Answer: (A) ECPFNG

Authenticity: Community Reported

Q6. If STOP is POTS, then LIVE is:

(A) EVIL ✓

(B) VILE

(C) LEVI

(D) EVLI

Answer: (A) EVIL

Authenticity: Community Reported

Q7. 'A + B' = A is father of B; 'A - B' = A is sister of B. In P + Q - R, P is R's:

(A) Brother

(B) Uncle

(C) Father ✓

(D) Grandfather

Answer: (C) Father

Authenticity: Community Reported

Q8. 'His mother is the only daughter of my mother,' says a woman about a man. She is his:

(A) Sister

(B) Aunt

(C) Mother ✓

(D) Grandmother

Answer: (C) Mother

Authenticity: Community Reported

Q9. Walk 4 km east then 3 km north. Distance from start:

(A) 7 km

(B) 5 km ✓

(C) 6 km

(D) 1 km

Answer: (B) 5 km

Authenticity: Community Reported

Q10. At sunrise you face the sun. Your left hand points:

(A) South

(B) North ✓

(C) East

(D) West

Answer: (B) North

Authenticity: Community Reported

Q11. Doctor : Hospital :: Teacher : ?

(A) Student

(B) Book

(C) School ✓

(D) Class

Answer: (C) School

Authenticity: Community Reported

Q12. Odd one out: 3, 5, 7, 9, 11.

(A) 5

(B) 7

(C) 9 ✓

(D) 11

Answer: (C) 9

Authenticity: Community Reported

Q13. In 12 hours the hands overlap how many times?

(A) 12

(B) 11 ✓

(C) 24

(D) 10

Answer: (B) 11

Authenticity: Community Reported

Q14. Aman is 7th from front and 11th from back in a queue. Total people:

(A) 16

(B) 17 ✓

(C) 18

(D) 19

Answer: (B) 17

Authenticity: Community Reported

Section 2 - Quantitative Ability

Q1. Find the next term: 1, 4, 9, 16, 25, ?

(A) 30

(B) 36 ✓

(C) 49

(D) 35

Answer: (B) 36

Authenticity: Frequently Repeated

Q2. Find the next term (primes): 2, 3, 5, 7, 11, ?

(A) 12

(B) 13 ✓

(C) 14

(D) 15

Answer: (B) 13

Authenticity: Frequently Repeated

Q3. If the price increases by 25%, by what percent must consumption fall to keep expenditure fixed?

(A) 25%

(B) 20% ✓

(C) 16.66%

(D) 10%

Answer: (B) 20%

Authenticity: Frequently Repeated

Q4. If 15% of x equals 20% of y, find x : y.

(A) 3 : 4

(B) 4 : 3 ✓

(C) 2 : 3

(D) 5 : 4

Answer: (B) 4 : 3

Authenticity: Frequently Repeated

Q5. An article sold at 20% loss fetched Rs. 480. Cost price:

(A) Rs. 600 ✓

(B) Rs. 576

(C) Rs. 500

(D) Rs. 620

Answer: (A) Rs. 600

Authenticity: Community Reported

Q6. A shopkeeper uses an 800 g weight for 1 kg. Profit percent:

(A) 20%

(B) 25% ✓

(C) 12.5%

(D) 16.66%

Answer: (B) 25%

Authenticity: Community Reported

Q7. A is twice as efficient as B; together 12 days. A alone finishes in:

(A) 18 ✓

(B) 24

(C) 16

(D) 20

Answer: (A) 18

Authenticity: Community Reported

Q8. A+B do it in 8 days, B+C in 12, A+B+C in 6. C alone:

(A) 24 days ✓

(B) 12 days

(C) 20 days

(D) 16 days

Answer: (A) 24 days

Authenticity: Community Reported

Q9. A 150 m train at 90 km/h crosses a pole in:

(A) 6 s ✓

(B) 8 s

(C) 10 s

(D) 5 s

Answer: (A) 6 s

Authenticity: Community Reported

Q10. Boat speed 10 km/h, stream 2 km/h. Downstream speed:

(A) 8 km/h

(B) 12 km/h ✓

(C) 10 km/h

(D) 6 km/h

Answer: (B) 12 km/h

Authenticity: Community Reported

Q11. If A:B = 2:3 and B:C = 4:5, then A:B:C is:

(A) 8:12:15 ✓

(B) 2:3:5

(C) 8:12:10

(D) 2:4:5

Answer: (A) 8:12:15

Authenticity: Community Reported

Q12. Two numbers in ratio 3:5 sum to 64. Larger:

(A) 24

(B) 40 ✓

(C) 35

(D) 45

Answer: (B) 40

Authenticity: Community Reported

Q13. Compound interest on Rs. 10000 at 10% for 2 years:

(A) Rs. 2000

(B) Rs. 2100 ✓

(C) Rs. 2200

(D) Rs. 1900

Answer: (B) Rs. 2100

Authenticity: Community Reported

Q14. CI on Rs. 8000 at 10% for 1 year, compounded half-yearly:

(A) Rs. 800

(B) Rs. 820 ✓

(C) Rs. 840

(D) Rs. 810

Answer: (B) Rs. 820

Authenticity: Research-Backed Company Style

Section 3 - Verbal Ability

Q1. Synonym of CANDID:

(A) Secretive

(B) Frank ✓

(C) Rude

(D) Shy

Answer: (B) Frank

Authenticity: Community Reported

Q2. Synonym of ENORMOUS:

(A) Tiny

(B) Huge ✓

(C) Narrow

(D) Weak

Answer: (B) Huge

Authenticity: Community Reported

Q3. Antonym of OPTIMIST:

(A) Realist

(B) Pessimist ✓

(C) Idealist

(D) Dreamer

Answer: (B) Pessimist

Authenticity: Community Reported

Q4. Antonym of VICTORY:

(A) Triumph

(B) Win

(C) Defeat ✓

(D) Success

Answer: (C) Defeat

Authenticity: Community Reported

Q5. Choose the correct sentence ('one of the'):

(A) One of the boys are absent.

(B) One of the boys is absent. ✓

(C) One of the boy is absent.

(D) One of boys are absent.

Answer: (B) One of the boys is absent.

Authenticity: Community Reported

Q6. Choose the correct sentence ('each of'):

(A) Each of the students have a book.

(B) Each of the students has a book. ✓

(C) Each of the student have a book.

(D) Each students has a book.

Answer: (B) Each of the students has a book.

Authenticity: Community Reported

Q7. She has been living here ____ 2010.

(A) for

(B) from

(C) since ✓

(D) by

Answer: (C) since

Authenticity: Community Reported

Q8. I am fond ____ music.

(A) for

(B) of ✓

(C) with

(D) to

Answer: (B) of

Authenticity: Community Reported

Q9. (Same passage) A stated limitation of renewables is:

(A) They cause pollution

(B) Output depends on weather ✓

(C) They are non-renewable

(D) They are unlimited

Answer: (B) Output depends on weather

Authenticity: Research-Backed Company Style

Q10. Order P,Q,R,S. P: He worked hard for months. Q: Rahul wanted to top the exam. R: Finally he succeeded. S: He made a strict timetable.

(A) Q S P R ✓

(B) Q P S R

(C) P Q S R

(D) S Q P R

Answer: (A) Q S P R

Authenticity: Research-Backed Company Style

Section 4 - Pseudocode / Technical

Q1. Which data structure follows FIFO?

(A) Stack

(B) Queue ✓

(C) Tree

(D) Graph

Answer: (B) Queue

Authenticity: Community Reported

Q2. Which BST traversal yields sorted order?

(A) Preorder

(B) Postorder

(C) Inorder ✓

(D) Level order

Answer: (C) Inorder

Authenticity: Community Reported

Q3. Which is NOT a linear data structure?

(A) Array

(B) Stack

(C) Queue

(D) Tree ✓

Answer: (D) Tree

Authenticity: Community Reported

Q4. for(i=0; i<3; i++) print(i); Output:

(A) 1 2 3

(B) 0 1 2 ✓

(C) 0 1 2 3

(D) 0 1

Answer: (B) 0 1 2

Authenticity: Community Reported

Q5. Postfix of (A + B) * C:

(A) AB+C* ✓

(B) ABC+*

(C) A+BC*

(D) AB*C+

Answer: (A) AB+C*

Authenticity: Community Reported

Q6. function g(n){ if(n==0) return 0; return n + g(n-1);} g(5):

(A) 10

(B) 15 ✓

(C) 20

(D) 12

Answer: (B) 15

Authenticity: Frequently Repeated

Section 5 - Coding

Problem 1: Maximum Subarray Sum (Kadane)

Difficulty: *Easy-Medium* | Pattern: *Array / DP* | Authenticity: *Frequently Repeated*

Problem Statement

Find the maximum sum of any contiguous non-empty subarray.

Input Format

Line 1: N. Line 2: N integers.

Output Format

Maximum contiguous subarray sum.

Constraints

$1 \leq N \leq 10^5$

Sample Input

```
9
-2 1 -3 4 -1 2 1 -5 4
```

Sample Output

```
6
```

Python Solution

```
def max_subarray(arr):
    best = cur = arr[0]
    for x in arr[1:]:
        cur = max(x, cur + x)
        best = max(best, cur)
    return best

print(max_subarray([-2, 1, -3, 4, -1, 2, 1, -5, 4]))
```

Step-by-Step Explanation

At each element extend the running sum or restart; track the best. Subarray [4,-1,2,1] gives 6.

Dry Run

cur peaks at 6 over [4,-1,2,1]; best=6.

Time Complexity: **O(N)** | Space Complexity: **O(1)**

Problem 2: Prime Pairs Summing to N (Goldbach)

Difficulty: **Medium** | Pattern: **Number Theory** | Authenticity: **Community Reported**

Problem Statement

Count unordered pairs of primes (p, q) with $p + q = n$ and $p \leq q$. (Relates to reported 'prime number pairs' problems.)

Input Format

A single integer n.

Output Format

Number of prime pairs.

Constraints

$1 \leq n \leq 10^5$

Sample Input

10

Sample Output

2

Python Solution

```
def prime_pairs(n):
    def is_prime(x):
        if x < 2:
            return False
        i = 2
        while i * i <= x:
            if x % i == 0:
                return False
            i += 1
        return True
    count = 0
    for p in range(2, n // 2 + 1):
        if is_prime(p) and is_prime(n - p):
            count += 1
    return count

print(prime_pairs(10))
```

Step-by-Step Explanation

For each p up to $n/2$, check if both p and $n-p$ are prime. For 10: (3,7) and (5,5).

Dry Run

2 valid prime pairs.

Time Complexity: **$O(n * \sqrt{n})$** | Space Complexity: **$O(1)$**



Full-Length Mock Test 2

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.